# Three-Prime Cubic CRT Checkpoint

Cross-pair expansion and the exact remaining obstruction

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## Contents


**1 Executive conclusion** **1**

**2 A. Frozen checkpoint** **2**

**3 B. Three-prime verdict** **2**

**4 C. Cross-pair expansion** **3**
4.1 C.1 The frozen unary shell family becomes expansive . . . . . 3
4.2 C.2 A polarized theorem for full antichain shells . . . . . 4
4.3 C.3 Weighted joins: all complete valuation-cell unions . . . . . 5

**5 D. Conditional proof of the full theorem** **7**
5.1 D.1 The missing polarized estimate . . . . . 7
5.2 D.2 The simultaneous-output color lemma . . . . . 7
5.3 D.3 Branch or renormalize . . . . . 9
5.4 D.4 Why the maximum is necessary . . . . . 10
5.5 D.5 The product-tree theorem for $C = 0$ . . . . . 10
5.6 D.6 What remains for nonzero $C$ . . . . . 12

**6 E. Why unary tensorization failure does not settle the binary problem** **13**

**7 F. Dependence of constants** **13**

**8 G. Implication for many primes and the global construction** **14**

**9 H. Exactly one next unresolved statement** **14**


## 1 Executive conclusion

Let

$$P_C(x, y) = (x - y)^3 + C(x + y), \qquad N = p^k q^\ell r^m,$$

where $p, q, r$ are fixed distinct odd primes. The assertion

$$|P_C(S,S)| \geq c_{p,q,r}\frac{|S|^3}{N^2} \tag{3CRT}$$

has **not** been proved or disproved in this checkpoint. No counterexample was found.

The strongest rigorous conclusions are:

1. The valuation-shell example that defeats unary two-prime tensorization is not a binary counterexample. Cross-pairs create distinct corner shells and restore more than the missing cubic factor.
2. For $C = 0$, the full three-prime theorem holds for **every** set $S$, with an explicit constant depending only on $p, q, r$. A four-tuple charging argument proves the missing product-tree energy inequality.
3. The full three-prime theorem follows from one precise polarized two-prime estimate. If, for $M = q^\ell r^m$,

$$|P_C(A,B)| \geq b_{q,r}\frac{|A||B|\max\{|A|,|B|\}}{M^2} \tag{BP2+}$$

holds uniformly in $\ell, m, C, A, B$, then (3CRT) holds with

$$c_{p,q,r} = \min\left\{\frac{c_{q,r}}{p^2}, \frac{b_{q,r}}{54}\right\}. \tag{1}$$

Here $c_{q,r}$ is the frozen two-prime same-set constant. Estimate (BP2+) is strictly stronger than the frozen theorem and remains open. Sparse occupancy and overlap inside two-prime valuation cells are the exact unresolved issue.

## 2 A. Frozen checkpoint

The five frozen results in the prompt were used as assumptions and were not re-proved:

- the single-prime restricted-image theorem;
- the two-prime CRT same-set theorem;
- failure of abstract same-set tensorization;
- the anchored unary cubic-slice theorem;
- failure of unary two-prime tensorization through incomparable valuation shells.

The previously closed prime-power, ball, coset, and standard-scaling branches were not reopened. No internet search or external retrieval was used, and no external verification is claimed.

## 3 B. Three-prime verdict

| The full three-prime theorem is unresolved here, but (BP2+) implies it. |
|---|

The known unary counterexample and all complete-shell variants are rigorously ruled out. In fact, the homogeneous case $C = 0$ is completely proved. Any counterexample to the uniform-in-$C$ statement must use the interaction between the cube and a nonzero, nonunit linear coefficient.

# 4 C. Cross-pair expansion

## 4.1 C.1 The frozen unary shell family becomes expansive

Put

$$R_q = \mathbb{Z}/q^\ell\mathbb{Z}, \qquad R_r = \mathbb{Z}/r^m\mathbb{Z}, \qquad M = q^\ell r^m.$$

For $3i < \ell$ and $3j < m$, define

$$V(i,j) = \{(x_q, x_r) : v_q(x_q) = i,\ v_r(x_r) = j\}.$$

Choose an antichain

$$i_1 < i_2 < \cdots < i_T, \qquad j_1 > j_2 > \cdots > j_T.$$

For $s < t$, unequal valuations give the exact difference shell

$$V(i_s, j_s) - V(i_t, j_t) = \{v_q = i_s,\ v_r = j_t\}. \tag{2}$$

Thus opposite endpoints control the two valuations of a cross-difference. This phenomenon is absent from an anchored unary image.

We use one standard theorem:

**Cyclicity of odd prime-power units.** If $\pi$ is an odd prime and $h \geq 1$, then $(\mathbb{Z}/\pi^h\mathbb{Z})^\times$ is cyclic. Hence the cube map on this group has kernel of order

$$\gcd(3, \varphi(\pi^h)) \leq 3.$$

It follows from (2) that

$$|P_0(V(i_s, j_s), V(i_t, j_t))| \geq \delta_{q,r} M q^{-3i_s} r^{-3j_t}, \qquad \delta_{q,r} = \frac{(1-q^{-1})(1-r^{-1})}{9}. \tag{3}$$

This output has valuation vector $(3i_s, 3j_t)$. Different unordered shell pairs therefore produce disjoint output shells.

For a balanced family, put $w_t = q^{-i_t} r^{-j_t}$ and suppose $H/R_0 \leq w_t \leq H$. The frozen unary calculation sees only scale $TH^3M$, whereas the cubic target is $T^3H^3M$. But the extreme binary cross-pair has size at least

$$\delta_{q,r} M (H/R_0)^3 \min(q,r)^{3(T-1)},$$

which eventually dominates $T^3H^3M$. Cross-pairs overcompensate for the unary loss exponentially.

## 4.2 C.2 A polarized theorem for full antichain shells

**Proposition 1.** With the antichain above, let

$$A = \bigcup_{s \in I} V(i_s, j_s), \qquad B = \bigcup_{t \in J} V(i_t, j_t)$$

for arbitrary $I, J \subseteq \{1, \dots, T\}$. Then

$$|P_0(A, B)| \geq b^{\text{shell}}_{q,r} \frac{|A||B| \max\{|A|, |B|\}}{M^2}, \tag{4}$$

where

$$b^{\text{shell}}_{q,r} = \frac{1}{18\kappa^2_{q,r} H^2_{q,r}}, \quad \kappa_{q,r} = (1 - q^{-1})(1 - r^{-1}), \quad H_{q,r} = 1 + \frac{2}{\min(q, r)^{3/2} - 1}. \tag{5}$$

**Proof.** Define

$$x_t = q^{-i_t}, \qquad y_t = r^{-j_t}, \qquad w_t = x_t y_t,$$

and

$$z_{st} = x_{\min(s,t)} y_{\max(s,t)}.$$

Then $|V(i_t, j_t)| = \kappa_{q,r} M w_t$. Equation (3), followed by a factor at most 2 for the two orientations of an unordered pair, gives

$$|P_0(A, B)| \geq \frac{\kappa_{q,r} M}{18} \sum_{s \in I, t \in J} z^3_{st}. \tag{6}$$

Let $\lambda = \min(q, r)$. Strict monotonicity yields

$$w_s \leq \lambda^{-|s-t|} z_{st}. \tag{7}$$

For fixed $t$, Holder's inequality with exponents 3 and 3/2 gives

$$\sum_{s \in I} z^3_{st} \geq H^{-2}_{q,r} \left( \sum_{s \in I} w_s \right)^3. \tag{8}$$

Let $X = \sum_{s \in I} w_s$ and $Y = \sum_{t \in J} w_t$. If $X \geq Y$, sum (8) over $t \in J$. Since each $w_t \leq 1$, one has $|J| \geq Y$, and therefore

$$\sum_{s \in I, t \in J} z^3_{st} \geq H^{-2}_{q,r} X^2 Y.$$

The case $Y \geq X$ is symmetric. Since

$$\frac{|A||B|\max(|A|,|B|)}{M^2} = \kappa_{q,r}^3 MXY \max(X,Y),$$

substitution in (6) proves (4). □

Thus the family responsible for unary failure satisfies the desired polarized binary estimate with an explicit constant.

### 4.3 C.3 Weighted joins: all complete valuation-cell unions

For $\lambda_1, \lambda_2, \lambda_3 > 1$, define

$$w(a) = \prod_{i=1}^{3} \lambda_i^{a_i}, \qquad a \vee b = (\max(a_1,b_1), \max(a_2,b_2), \max(a_3,b_3)).$$

For finite $I \subseteq \mathbb{Z}_{\geq 0}^3$, put $J = I \vee I$ and

$$D = \prod_{i=1}^{3} (1-\lambda_i^{-1})^{-1}, \qquad C_\lambda = D^3 \left(1 + 6\sum_{i=1}^{3} \frac{1}{\lambda_i^3 - 1}\right). \tag{9}$$

**Lemma 2 (weighted join lemma).**

$$\left(\sum_{a \in I} w(a)\right)^3 \leq C_\lambda \sum_{z \in J} w(z)^3. \tag{10}$$

**Proof.** Expand over ordered triples $(a,b,c) \in I^3$.

If one point, say $c$, lies below $a \vee b$, charge the triple to $z = a \vee b$. Since

$$\sum_{x \leq z} w(x) \leq D w(z),$$

the mass charged to fixed $z$ is at most $D^3 w(z)^3$.

Otherwise each point is the unique maximum in a distinct coordinate. Relabel so that $a, b, c$ have unique maxima in coordinates $1, 2, 3$. Choose the heaviest pair join; suppose it is $z = a \vee b$. Write $h = c_3 - z_3 \geq 1$ and

$$\Delta_1 = (z_1 - \max(b_1, c_1)) \log \lambda_1, \qquad \Delta_2 = (z_2 - \max(a_2, c_2)) \log \lambda_2.$$

Comparing the three pair-join weights gives

$$\Delta_1, \Delta_2 \geq h \log \lambda_3. \tag{11}$$

In the downset sum, $a$ loses at least $\Delta_2$, $b$ loses at least $\Delta_1$, and $c$ gains $h \log \lambda_3$ in coordinate 3 but loses at least $\Delta_1 + \Delta_2$ in coordinates $1, 2$. By (11), each of the three resulting factors is at most $\lambda_3^{-h}$. Thus the mass with fixed $(z, h)$ is at most

$$D^3 w(z)^3 \lambda_3^{-3h}.$$

There are 6 assignments of the unique maxima. The omitted coordinate may be any of the three, and

$$\sum_{h \geq 1} \lambda_j^{-3h} = \frac{1}{\lambda_j^3 - 1}.$$

Adding both cases proves (10). $\square$

Now take

$$N = p^{3K_p} q^{3K_q} r^{3K_r}, \qquad B = N^{2/3}.$$

For $s \in \{p, q, r\}$, define

$$E_s(0) = s^{K_s}(\mathbb{Z}/s^{3K_s}\mathbb{Z})$$

and, for $1 \leq a \leq K_s$,

$$E_s(a) = \{x : v_s(x) = K_s - a\}.$$

For $a = (a_p, a_q, a_r)$ put

$$E(a) = E_p(a_p) \times E_q(a_q) \times E_r(a_r), \qquad w(a) = p^{a_p} q^{a_q} r^{a_r}.$$

Then

$$|E(a)| \leq B w(a). \tag{12}$$

Complete-cell subtraction satisfies

$$E(a) - E(b) \supseteq E(a \vee b). \tag{13}$$

If two local valuations differ, the lower valuation dominates. If they agree, every unit residue is a difference of two units; over an odd residue field, choose a nonzero residue different from the negative of the prescribed unit.

Cyclicity of odd prime-power units gives

$$|\{u^3 : u \in E(z)\}| \geq \kappa^{\mathrm{cube}}_{p,q,r} w(z)^3, \qquad \kappa^{\mathrm{cube}}_{p,q,r} = \prod_{s\in\{p,q,r\}} \frac{1-s^{-1}}{3}. \tag{14}$$

When $z_s = 0$, the local cube image is $\{0\}$, and the factor in (14) is still valid. Distinct $z$ have distinct cube-output valuation vectors, so their images are disjoint.

Therefore every complete-cell union

$$S = \bigcup_{a\in I} E(a)$$

satisfies the unconditional estimate

$$\boxed{|P_0(S,S)| \geq \frac{\kappa^{\mathrm{cube}}_{p,q,r}}{C_{(p,q,r)}} \frac{|S|^3}{N^2}.} \tag{15}$$

This includes zero-centered chains, antichains, and arbitrary valuation-Cantor unions. It does not include sparse subsets inside the cells.

# 5 D. Conditional proof of the full theorem

## 5.1 D.1 The missing polarized estimate

Let

$$R_{q,r} = \mathbb{Z}/q^\ell\mathbb{Z} \times \mathbb{Z}/r^m\mathbb{Z}, \qquad M = q^\ell r^m.$$

Hypothesis (BP2+) asserts that there is $b_{q,r} > 0$, depending only on $q, r$, such that for every $\ell, m, C$ and all nonempty $A, B \subseteq R_{q,r}$,

$$|P_C(A,B)| \geq b_{q,r} \frac{|A||B| \max\{|A|,|B|\}}{M^2}. \tag{16}$$

Taking $A = B$ has the shape of the frozen theorem, but the same-set theorem does not imply (16).

## 5.2 D.2 The simultaneous-output color lemma

Put $P = p^k$ and $N = PM$. Assume $p^2 \mid C_p$, where $C_p$ is the $p$-primary component of $C$. Let $U, V \subseteq \mathbb{Z}/N\mathbb{Z}$ have disjoint first $p$-digit supports. Decompose into exact $p$-coordinate fibers:

$$U = \bigcup_i \{i\} \times U_i, \qquad V = \bigcup_j \{j\} \times V_j,$$

where $U_i, V_j \subseteq R_{q,r}$. Write $a_i = |U_i|$, $b_j = |V_j|$, and let $I, J$ denote the numbers of occupied exact fibers.

Color an edge $(i,j)$ by its $p$-adic output

$$h(i,j) = (i-j)^3 + C_p(i+j) \pmod{p^k}.$$

For fixed $j$, a color has at most three preimages $i$. Modulo $p$, the equation is a cube equation, so it has at most three roots. Within one nonzero-difference residue class,

$$h(i,j) - h(i',j) = (i-i')\left((i-j)^2 + (i-j)(i'-j) + (i'-j)^2 + C_p\right). \tag{17}$$

For $p \neq 3$, the bracket is a unit, so the lift is injective. For $p = 3$, the bracket has valuation exactly 1, giving at most three lifts. The same argument applies in columns. Hence each color class has at most

$$3\min(I,J) \tag{18}$$

edges.

This controls actual simultaneous CRT outputs. Under (BP2+),

$$\begin{aligned} |P_C(U,V)| &\geq \frac{1}{3\min(I,J)} \sum_{i,j} |P_C(U_i,V_j)| \\ &\geq \frac{b_{q,r}}{3M^2\min(I,J)} \sum_{i,j} a_i b_j \max(a_i,b_j). \end{aligned} \tag{19}$$

Let $u = |U|$ and $v = |V|$. Finite Cauchy-Schwarz gives

$$\sum_{i,j} a_i b_j \max(a_i,b_j) \geq \frac{u^2 v}{I}, \qquad \sum_{i,j} a_i b_j \max(a_i,b_j) \geq \frac{uv^2}{J}.$$

Consequently,

$$\frac{\sum_{i,j} a_i b_j \max(a_i,b_j)}{\min(I,J)} \geq \frac{uv\max(u,v)}{IJ}.$$

Since $I, J \leq P$, we obtain:

**Lemma 3 (cross-branch lifting).**

$$\boxed{|P_C(U,V)| \geq \frac{b_{q,r}}{3}\frac{|U||V|\max\{|U|,|V|\}}{N^2}.} \tag{20}$$

### 5.3 D.3 Branch or renormalize

We prove (3CRT) conditionally by induction on $k$.

**Low valuation of $C_p$.** Suppose $v_p(C_p) \le 1$. For every exact $p$-coordinate $a$, let $S_a \subseteq R_{q,r}$ be the fiber and put $n_a = |S_a|$. Pairs within this fiber have $p$-output $2C_p a$. The map $a \mapsto 2C_p a$ has multiplicity at most $p$. The frozen two-prime theorem and Holder give

$$\begin{aligned} |P_C(S,S)| &\ge \frac{c_{q,r}}{pM^2} \sum_a n_a^3 \\ &\ge \frac{c_{q,r}}{p} \frac{|S|^3}{N^2}. \end{aligned} \tag{21}$$

**Base exponents.** If $k \le 2$, project $S$ to $R_{q,r}$. Its projection has size at least $|S|/P$, and the frozen theorem gives

$$|P_C(S,S)| \ge \frac{c_{q,r}}{P} \frac{|S|^3}{N^2} \ge \frac{c_{q,r}}{p^2} \frac{|S|^3}{N^2}. \tag{22}$$

**Renormalization.** Suppose $p^2 \mid C_p$ and $S$ lies in one first $p$-digit. In the $p$-coordinate write $x = a + pX$, $y = a + pY$. Then

$$P_{C_p}(a + pX, a + pY) = 2aC_p + p^3 P_{C_p/p^2}(X,Y). \tag{23}$$

Projecting $X, Y$ modulo $p^{k-3}$ loses at most $p^2$ in input size. The output map in (23) is injective, while the $q, r$ coordinates are unchanged. The target is exactly invariant:

$$\frac{(|S|/p^2)^3}{(p^{k-3}M)^2} = \frac{|S|^3}{N^2}. \tag{24}$$

**Branching.** Partition $S$ into first $p$-digit classes.

If no class contains more than $2|S|/3$, split the classes into unions $U, V$, each of size between $|S|/3$ and $2|S|/3$. Lemma 3 safely gives

$$|P_C(S,S)| \ge \frac{b_{q,r}}{54} \frac{|S|^3}{N^2}. \tag{25}$$

If one class $U$ is dominant, let $u = |U| > 2|S|/3$, let $V = S \setminus U$, and put $v = |V|$. The self-image $P_C(U,U)$ is $0 \pmod p$, while the cross-image $P_C(U,V)$ is nonzero modulo $p$; the two images are disjoint. Renormalization and induction give $cu^3/N^2$ self-outputs, while Lemma 3 gives at least

$$\frac{b_{q,r}}{3} \frac{u^2 v}{N^2}$$

cross-outputs. Since $v < u/2$,

$$(u+v)^3 - u^3 = 3u^2v + 3uv^2 + v^3 \le 5u^2v.$$

The induction closes when $c \le b_{q,r}/15$. Combining (21), (22), and (25) proves the conditional constant (1).

### 5.4 D.4 Why the maximum is necessary

If a dominant branch has mass $u$ and its remainder has mass $v \ll u$, the cubic target lost by retaining only the dominant branch is

$$(u+v)^3 - u^3 \asymp u^2 v.$$

Thus a cross estimate must also have order $u^2v/N^2$. A bound with $|A||B|\min(|A|,|B|)$ gives only $uv^2/N^2$ and cannot close a depth-uniform induction.

### 5.5 D.5 The product-tree theorem for $C = 0$

For $d \in \mathbb{Z}/M\mathbb{Z}$, let $\mathrm{ord}(d)$ be its additive order. Under CRT,

$$\frac{\mathrm{ord}(d)}{M} = q^{-\min(v_q(d_q),\ell)} r^{-\min(v_r(d_r),m)}. \tag{26}$$

The following convolution lemma is the new mechanism that controls arbitrary sparse cells.

**Lemma 4 (triple convolution of low-order differences).** For a divisor $H \mid M$, put

$$D_H = \{d \in \mathbb{Z}/M\mathbb{Z} : \mathrm{ord}(d) \le H\}.$$

If $\mathrm{ord}(d_0) = H$, then

$$\#\{(u,v,w) \in D_H^3 : u+v+w = d_0\} \le C_{q,r}^{\mathrm{conv}} H^2, \tag{27}$$

where, with

$$A_q = (1-q^{-1})^{-1}, \qquad A_r = (1-r^{-1})^{-1},$$

one may take

$$C_{q,r}^{\mathrm{conv}} = 6(A_q A_r^2 + A_q^2 A_r). \tag{28}$$

**Proof.** For an exponent pair $\alpha = (i,j)$, let $K_\alpha$ be the unique subgroup of $\mathbb{Z}/M\mathbb{Z}$ of order $q^i r^j$. Cover $D_H$ by the subgroups $K_{\alpha_s}$ for which $q^{i_s} r^{j_s} \le H$ is maximal under divisibility. Order them so that

$$i_1 < \cdots < i_T, \qquad j_1 > \cdots > j_T.$$

Write $H = q^a r^b$. Since $H$ itself is a maximal divisor, some distinguished index $t$ has $\alpha_t = (a,b)$.

For $i \le j \le k$, the sum $K_{\alpha_i} + K_{\alpha_j} + K_{\alpha_k}$ has order $q^{i_k} r^{j_i}$. If it contains $d_0$, then $i \le t \le k$. The addition homomorphism from the product of the three subgroups onto their sum has constant fiber size, so the number of representations contributed by this subgroup triple is

$$\frac{|K_{\alpha_i}||K_{\alpha_j}||K_{\alpha_k}|}{|K_{\alpha_i} + K_{\alpha_j} + K_{\alpha_k}|} = |K_{\alpha_j}| q^{i_i} r^{j_k}.$$

The exponent pairs form a strict antichain, hence

$$i_j - i_i \ge j - i, \qquad j_j - j_k \ge k - j.$$

Since $|K_{\alpha_j}| \le H$, the last representation count is at most

$$H^2 q^{-(j-i)} r^{-(k-j)}.$$

For $j \le t$, summing first in $i, k$ and then in $j$ costs at most $A_q A_r^2 H^2$. For $j > t$, it costs at most $A_q^2 A_r H^2$. Multiplying by 6 for the possible orderings proves (27). Covering by maximal subgroups may count elements more than once, which only enlarges this upper bound. □

**Proposition 5 (product-tree energy).** For all $A, B \subseteq \mathbb{Z}/M\mathbb{Z}$,

$$\boxed{\sum_{x \in A, y \in B} \operatorname{ord}(x-y)^2 \ge \frac{|A|^2 |B|^2}{4 C_{q,r}^{\mathrm{conv}}}.} \tag{29}$$

**Proof.** Consider an ordered quadruple $(x_1, x_2, y_1, y_2) \in A^2 \times B^2$. Choose canonically one of its four cross-edges $x_i - y_j$ having maximal additive order $H$, and suppose the chosen edge is $d = x_1 - y_1$. The three signed differences

$$u = x_2 - y_1, \qquad v = x_1 - y_2, \qquad w = -(x_2 - y_2)$$

belong to $D_H$ and satisfy

$$u + v + w = d.$$

For a fixed chosen edge and a fixed one of its four possible positions, the triple $(u, v, w)$ determines the other two vertices. Lemma 4 therefore allows at most $C_{q,r}^{\mathrm{conv}} H^2$ quadruples. Summing over the four positions proves (29). □

Combining (26) and (29) proves the product-tree inequality

$$\sum_{x \in A, y \in B} q^{-2\min(v_q(x_q - y_q), \ell)} r^{-2\min(v_r(x_r - y_r), m)} \ge \frac{|A|^2 |B|^2}{4 C_{q,r}^{\mathrm{conv}} M^2}. \tag{30}$$

It remains to pass from weighted differences to cube outputs. For one odd prime $\pi$ and every $z \pmod{\pi^h}$,

$$\sum_{u^3=z} \pi^{-2\min(v_\pi(u),h)} \le 3. \tag{31}$$

For $z \neq 0$, all roots have one valuation $s$, there are at most $3\pi^{2s}$ roots, and each has weight $\pi^{-2s}$. For $z = 0$, direct summation over $s \ge \lceil h/3 \rceil$ is a geometric series smaller than 3. CRT makes the two-prime bound in (31) at most 9.

Every difference has at most $\min(|A|, |B|)$ representations. Grouping the weighted pair sum in (30) by the cube output therefore gives:

**Corollary 6 (homogeneous polarized two-prime theorem).**

$$\boxed{|P_0(A,B)| \ge b^0_{q,r} \frac{|A||B| \max\{|A|,|B|\}}{M^2}, \qquad b^0_{q,r} = \frac{1}{36 C^{\mathrm{conv}}_{q,r}}.} \tag{32}$$

Thus (BP2+) is proved completely when $C = 0$. Substitution of (32) into the conditional induction proves the full homogeneous three-prime theorem:

$$\boxed{|P_0(S,S)| \ge c^0_{p,q,r} \frac{|S|^3}{N^2} \quad \text{for every } S \subseteq \mathbb{Z}/N\mathbb{Z},} \tag{33}$$

where

$$c^0_{p,q,r} = b^0_{q,r} \min\left\{\frac{1}{p^2}, \frac{1}{54}\right\} = \frac{1}{36 C^{\mathrm{conv}}_{q,r}} \min\left\{\frac{1}{p^2}, \frac{1}{54}\right\}. \tag{34}$$

This is stronger than the complete-cell theorem (15): it permits arbitrary CRT correlations and arbitrary sparse occupancy.

### 5.6 D.6 What remains for nonzero $C$

There is a useful endpoint estimate. In $(u, v) = (x - y, x + y)$ coordinates, which are invertible for odd moduli,

$$P_C = u^3 + Cv.$$

Let $g = \gcd(C, M)$. For fixed output and fixed $u$, the congruence in $v$ has at most $g$ solutions. Hence every full-ring fiber has at most $Mg$ points, and every restricted pair satisfies

$$|P_C(A,B)| \ge \frac{|A||B|}{Mg}. \tag{35}$$

In particular, (BP2+) holds with constant 1 when $C$ is a unit modulo $M$, since $\max(|A|, |B|) \le M$. More generally, (35) is sufficient whenever

$$\max(|A|, |B|) \le \frac{M}{g}.$$

The unresolved regime is therefore intermediate singular reduction: $1 < g < M$ together with dense, sparsely correlated sets. The homogeneous charging proof uses that the output is determined by the cube of one difference; it does not directly control the additional correlated variable $x + y$.

# 6 E. Why unary tensorization failure does not settle the binary problem

The unary counterexample counts only anchored images. Binary differences use both moving inputs. Every pair of incomparable shells creates the corner of its valuation rectangle; distant shells create exponentially larger corners, and the output valuation vector prevents overlap.

This is not numerical evidence: Proposition 1 proves the exact polarized estimate on the counterexample family, and Lemma 2 proves the same-set cubic bound for every union of complete three-prime valuation cells.

Raw collision energy is also inadequate. On $\mathbb{Z}/p^{3K}\mathbb{Z}$, the value 0 has $p^{2K}$ cube roots. On the full set, this singular fiber alone makes ordinary cube-collision energy exceed the scale needed for a direct Cauchy-Schwarz proof by a factor $p^K$. Any successful energy proof must weight or prune singular fibers.

# 7 F. Dependence of constants

The unconditional constants proved here are explicit:

- Proposition 1 has the constant in (5).
- Complete three-prime valuation-cell unions with $C = 0$ have constant

$$\frac{\kappa^{\mathrm{cube}}_{p,q,r}}{C_{(p,q,r)}}.$$

- For arbitrary two-prime sets in the homogeneous polarized theorem,

$$b^0_{q,r} = \frac{1}{216(A_q A_r^2 + A_q^2 A_r)}.$$

- For arbitrary three-prime sets in the homogeneous theorem, one may take

$$c^0_{p,q,r} = \frac{1}{216(A_q A_r^2 + A_q^2 A_r)} \min\left\{\frac{1}{p^2}, \frac{1}{54}\right\}.$$

Conditionally on (BP2+), the full theorem has

$$c_{p,q,r} = \min\left\{\frac{c_{q,r}}{p^2}, \frac{b_{q,r}}{54}\right\}.$$

One may cyclically choose the distinguished prime and take the best resulting bound. Every displayed constant is independent of $k, \ell, m, C, S$. No unconditional constant for arbitrary $S$ is claimed.

## 8 G. Implication for many primes and the global construction

This checkpoint closes the entire three-prime local restricted-input problem for the homogeneous map $(x-y)^3$, not merely the shell example or complete valuation cells.

It does not close the uniform problem for $(x-y)^3+C(x+y)$ with intermediate singular $C$. Therefore it does not yet close the many-prime cubic split-prime method or construct real sets for the nonhomogeneous polynomial.

Even if (BP2+) is proved, naive iteration over

$$N=\prod_{i=1}^{t} q_i^{k_i}$$

would repeatedly pay a branching/coloring constant comparable to 54, in addition to polarized constants inherited from the remaining coordinates. This produces at best exponential deterioration in $t$ unless a stronger uniform multilinear statement is found. Such decay can cancel the local amplification required by the global number-field construction. A theorem for fixed $t$ is therefore not by itself a quantitative solution to the growing-prime problem.

## 9 H. Exactly one next unresolved statement

Fix distinct odd primes $q,r$. Prove or disprove the following statement in the only remaining regime. There is $b_{q,r}>0$ such that

$$\boxed{|P_C(A,B)|\geq b_{q,r}\frac{|A||B|\max\{|A|,|B|\}}{(q^\ell r^m)^2}}$$

uniformly for all $\ell,m\geq 0$, all $C$, and all nonempty $A,B$ satisfying

$$1<\gcd(C,q^\ell r^m)<q^\ell r^m,\qquad \max(|A|,|B|)>\frac{q^\ell r^m}{\gcd(C,q^\ell r^m)}.$$

The excluded endpoint cases are already proved: $C=0$ by the triple-convolution charging theorem, and unit $C$ or smaller sets by the full-fiber bound (35). A positive answer to this one statement completes (BP2+) and hence proves (3CRT).